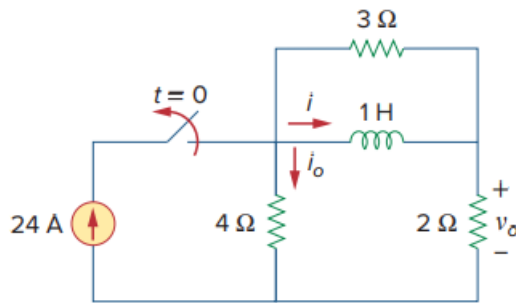


Problem

Determine i , i_o , and v_o for all t in the circuit shown in Fig. 7.22. Assume that the switch was closed for a long time. It should be noted that opening a switch in series with an ideal current source creates an infinite voltage at the current source terminals. Clearly this is impossible. For the purposes of problem solving, we can place a shunt resistor in parallel with the source (which now makes it a voltage source in series with a resistor). In more practical circuits, devices that act like current sources are, for the most part, electronic circuits. These circuits will allow the source to act like an ideal current source over its operating range but voltage-limit it when the load resistor becomes too large (as in an open circuit).

Fig. 7.22:



Step-by-step solution

Step 1 of 11

Refer Figure 7.22 from the textbook.

Switch is closed for a long time before switch is opened at $t = 0$

For $t < 0$, inductor is short circuited and the equivalent circuit is shown in Figure 1.

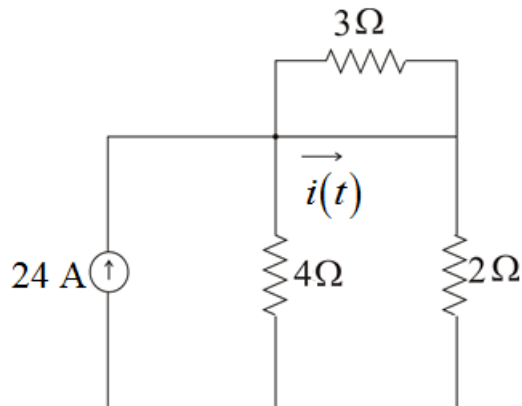


Figure 1

Step 2 of 11

There is a short circuit across $3\ \Omega$ resistor, so remove $3\ \Omega$ resistor. The resultant circuit is shown in Figure 2.

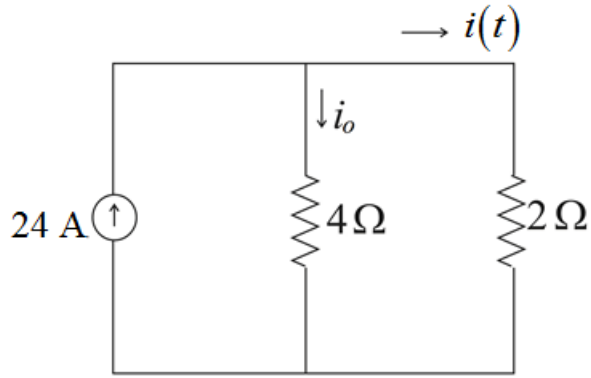


Figure 2

Step 3 of 11

Apply current division rule to Figure 2.

$$i(t) = 24 \left(\frac{4}{2+4} \right)$$

$$= 16\text{ A}$$

Since current through inductor cannot change instantaneously, the current is,

$$i(0) = i(0^+) = 12\text{ A}$$

Apply current division rule.

$$i_o(t) = 24 \left(\frac{2}{2+4} \right)$$

$$= 8\text{ A} \quad \text{for } t < 0$$

v_o is the voltage across $2\ \Omega$ resistor.

$$v_o(t) = 2i(t)$$

$$= 2(16)$$

$$= 32\text{ V} \quad \text{for } t < 0$$

Step 4 of 11

For $t > 0$, the switch is opened and the equivalent circuit is shown in Figure 3.

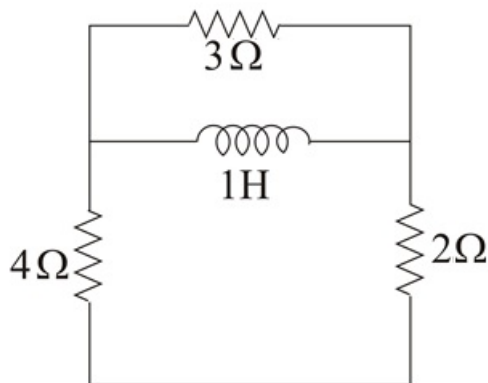


Figure 3

Step 5 of 11

Find the equivalent resistance across inductor terminals as shown in Figure 4.

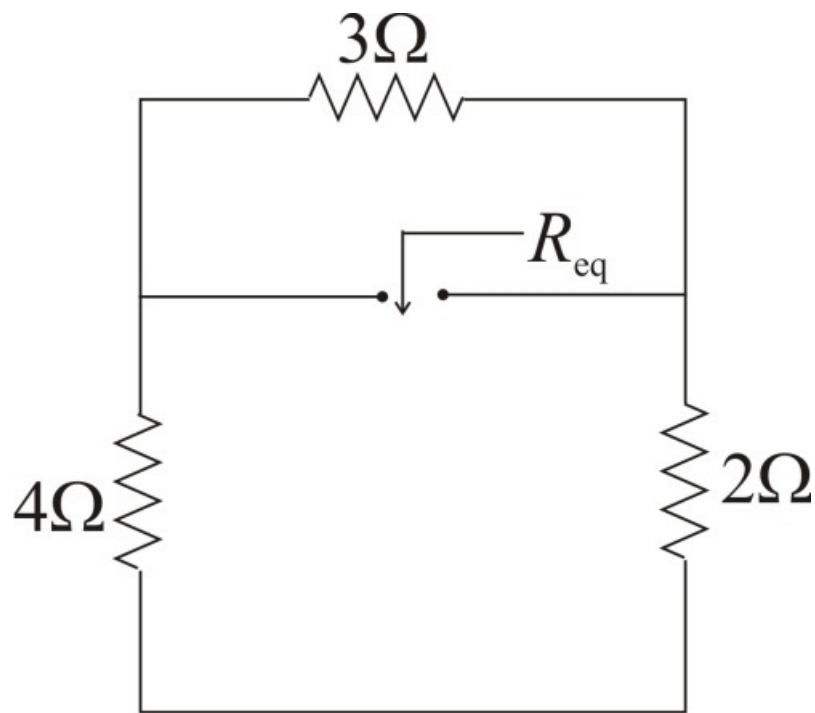


Figure 4

Step 6 of 11

The 4Ω and 2Ω resistors are in series. So draw the equivalent circuit as shown in Figure 5.

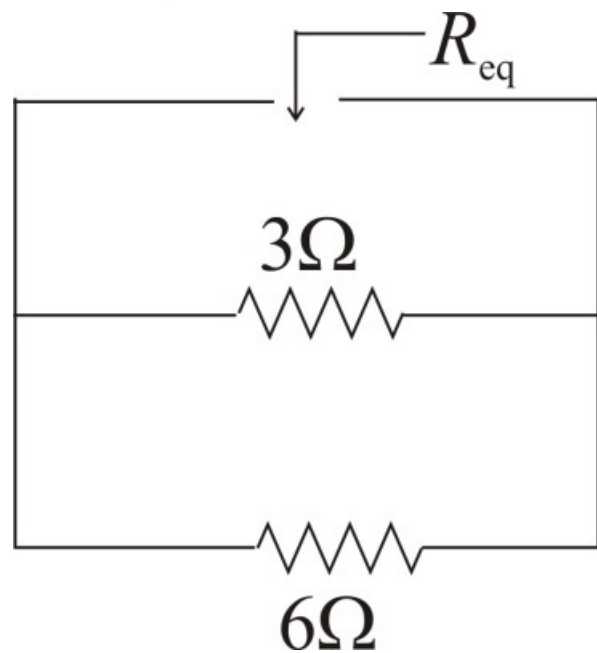


Figure 5

Step 7 of 11

Now, calculate the equivalent resistance from Figure 5.

$$\begin{aligned} R_{\text{eq}} &= \frac{(6)(3)}{6+3} \\ &= 2 \, \Omega \end{aligned}$$

The modified circuit is shown in Figure 6.

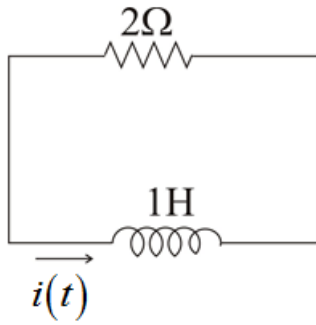


Figure 6

Step 8 of 11

Calculate the time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{\text{eq}}} \\ &= \frac{1}{2} \end{aligned}$$

The inductor current for $t > 0$ is,

$$\begin{aligned} i(t) &= i(0)e^{-\frac{t}{\tau}} \\ &= 16e^{-\left(\frac{t}{\frac{1}{2}}\right)} \\ &= 16e^{-2t} \text{ A for } t > 0 \end{aligned}$$

Thus, the current i for all t is,

$$i = \begin{cases} 16 \text{ A} & \text{for } t < 0 \\ 16e^{-2t} \text{ A} & \text{for } t \geq 0 \end{cases}$$

Step 9 of 11

For $t > 0$, the switch is opened and the equivalent circuit is shown in Figure 7.

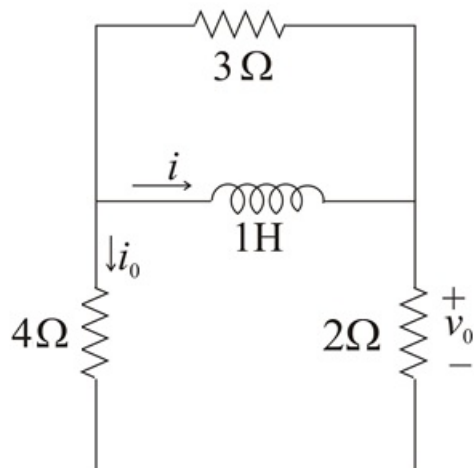


Figure 7

Apply current division rule to Figure 7.

$$\begin{aligned}
 i_o(t) &= -i(t) \left(\frac{3}{4+2+3} \right) \\
 &= -i(t) \left(\frac{1}{3} \right) \\
 &= -16e^{-2t} \left(\frac{1}{3} \right) \\
 &= -5.333e^{-2t} \text{ A for } t > 0
 \end{aligned}$$

Thus, the current i_o for all t is,

$$i_o = \begin{cases} 8 \text{ A}, & t < 0 \\ -5.333e^{-2t} \text{ A}, & t > 0 \end{cases}$$

Calculate voltage v_o for $t > 0$.

$$\begin{aligned}
 v_o(t) &= -2i_o(t) \\
 &= -2(-5.333e^{-2t}) \\
 &= 10.667e^{-2t} \text{ for } t > 0
 \end{aligned}$$

Thus, the voltage v_o for all t is,

$$v_o = \begin{cases} 32 \text{ V}, & t < 0 \\ 10.667e^{-2t} \text{ V}, & t > 0 \end{cases}$$